

6.1.94 WBGT index (Wet Bulb Globe Temperature index) - means a number which characterises the thermal conditions in the environment to which that number applies, calculated from dry bulb (outdoors), wet bulb and globe thermometers temperature readings. Refer Environmental Regulations for Workplaces of the Act.

6.1.95 Warning sign (danger) - means a sign in approved form calling attention to the danger of approach to, or interference with the plant or equipment to which it is attached.

6.1.96 Work / working - means all human activities in connection with plant, excluding operating activities and non-dangerous activities performed on the external parts of plant and which cannot affect the health and safety of workers or the safe operation of the plant.

6.1.97 Worker - means a person who has been allocated work to be carried out under a Permit to Work; this may also be the Authorised Supervisor.

6.1.98 Workers Register - means a list of workers that is kept current on a register for the allocation of work under the control of a specific Permit to Work.

KEY LEARNING POINTS
QUESTIONS AND ANSWERS
THEORY

- 1) State what is by definition "work".
 - a) All human activities in connection with the plant.**
 - b) Excluding operating activities
 - c) Non dangerous activities performed on the external parts of the plant which cannot affect the health and safety of workers or the safe operation of the plant.
 - d) An activity that does not require supervision by the Responsible Person
 - e) Activities carried out by a skilled person.
- 2) State what is defined as "plant".
 - a) Portable lifting equipment.
 - b) Plant and apparatus covered by the operating regulations for high voltage systems.
 - c) Any structures, machinery, apparatus, or equipment.**
 - d) Mobile lifting equipment and workshop equipment.
 - e) Power circuits.
- 3) Define the "clean hands approach".
 - a) Washing hands before removing fuses.
 - b) Having both hands clean and free to carry out isolations safely.**
 - c) Having hands clean before completing the Permit to Work documentation.
- 4) State the requirements for entry into a confined space.
 - a) A Gas Test Certificate must be issued.**
 - b) The workers must be secured by a safety harness.
 - c) Adequate breathing apparatus must be provided.
 - d) Gas monitoring must be maintained.**
 - e) Adequate ventilation must be provided and maintained.**
- 5) Define a Skilled person.
 - a) A person who has been trained, and declared competent in writing**
 - b) A fitter or electrician.
 - c) A person authorised to supervise work.
 - d) Has adequate knowledge for the task at hand**

- 6) Who is the "person in charge of the operation of the plant"
- a) The Operating Manager.
 - b) The Senior Shift Supervisor on duty at that particular time.
 - c) The person delegated in writing to be in charge of the operation of the plant.**
- 7) Who may carry out adjustments on plant in operation under a Permit to Work?
- a) The Responsible Person.
 - b) Any qualified artisan.
 - c) A Skilled person under the direct supervision of a Responsible Person**
 - d) A Responsible Person under the direct supervision of another Responsible Person.**
 - e) If an RP carries out the adjustments another RP or Skilled person must be in attendance
- 8) When is a Permit to Work deemed to be in force.
- a) When the Responsible Person signs the application.
 - b) When the Appointed Person issues the permit.
 - c) When the Responsible Person signs acceptance.**
- 9) An Operating activity is defined as.
- a) All the work carried out by the operator.
 - b) The opening and closing of breakers/ isolators and valves/ dampers for the control of the plant and to effect isolations.**
 - c) Any work carried out under the instructions of the shift supervisor.
 - d) Other activities involved with the process of electricity generation which are done in accordance with formulated safe procedures.**
- 10) What is the process for gaining access to keys on a Key safe or removing safety locks from the plant if the keys are not available?
- a) The Appointed Person cuts off the locks.
 - b) The Appointed Person must get permission from the relevant head of department/function, who must witness the process of lock removal.**
 - c) The Appointed Person and head of shift must record the activity in their log.**
 - d) The Power Station Manager must give permission and witness the activity.
- 11) When does new plant come under the control of the plant safety regulations.
- a) When the plant is taken over.
 - b) When the plant is energised for the first time.
 - c) As soon as the Safety clearance certificate is issued.**
 - d) When commissioning is completed.

- 12) What does the regulation state if there is doubt concerning the procedure to be adopted with respect to the application of the plant safety regulations.
- a) Ask your supervisor what to do.
 - b) Do whatever you think is right.
 - c) Consult the regulations for guidance**
 - d) Ask other responsible or Appointed Persons what they do.
- 13) The responsibilities of the Responsible Person are.
- a) Isolating the plant.
 - b) Acceptance and clearance of permits to work.**
 - c) Ensures work can be executed taking health and safety precautions into account and within the terms of these regulations.**
 - d) Removing safety locks from the plant.
 - e) Energising the plant for test purposes.
 - f) Maintaining a Workers' Register.**
 - g) Test running of plant.**
- 14) What must be done in order to test run plant subject to an unrevoked Permit to Work.
- a) Clear or suspend all relevant permits to work.**
 - b) Issue a Safety clearance certificate.
 - c) Remove all plant isolations.
 - d) Ensure that the Workers' Register(s) is withdraw.**
 - e) Issue a Sanction for Test.**
- 15) What extra precautions must be taken when entering a confined space where there is a possibility of dangerous substances being present and effective isolation and ventilation are not possible?
- a) The workers must not enter until a gas test has been done.
 - b) A Rescuer must remain outside and maintain communication with the workers inside.**
 - c) All workers must be medically fit before entering.
 - d) All workers must wear breathing apparatus and be secured by a safety harness and line.**
 - e) All workers must have a torch.
- 16) Can a person who is authorised as an Appointed and Responsible Person, simultaneously exercise both authorisations on the same Permit to Work.
- a) Yes.
 - b) Only with the permission of the head of department / function.
 - c) Yes, if so authorised.
 - d) Yes always.
 - e) No**

- 17) When is a Permit to Work required.
- a) When maintenance must be done.
 - b) When work requiring an isolation must be done.
 - c) Whenever work is to be done on plant.**
 - d) For any activity on the plant.
- 18) Under what conditions can a non-return valve be used as a point of isolation.
- a) If it is of the screw-down type.**
 - b) If the head of department/ function sanctions the work.
 - c) If the work is essential, the non-return valve is the only point of isolation, the plant cannot be shut down and an approved local procedure is available.**
 - d) If the system pressure is less than 50 kPa.**
 - e) If the system pressure is less than 600 kPa.
- 19) What is the definition of danger or dangerous.
- a) An area demarcated by danger signs.
 - b) A condition that is unsafe.
 - c) A condition or substance that constitutes a risk of personal injury, impairment of health, death or plant damage**
 - d) A condition that may result in an explosion.
- 20) The responsibilities of the Appointed Person are.
- a) Clearing of permits to work.
 - b) Test running of plant.**
 - c) Checking that isolations are correctly implemented.
 - d) Correctly isolating plant and ensuring it is safe to work on, taking health and safety precautions into account and within the terms of these regulations.**
 - e) The issue and revocation of permits to work.**
 - f) Reinstating plant after revoking a Permit to Work.**
- 21) What is the definition of a safety lock?
- a) A unique series of locks with a master key issued to authorised persons only.
 - b) A series of locks each having a unique and duplicate key.
 - c) A lock used for locking isolations for which only a single unique key is available.**

- 22) Under what circumstances can an outage Permit to Work be issued.
- a) No special circumstances.
 - b) Any outages
 - c) **Only on an outage declared by the general manager / project manager, where a designated manager has been Appointed.**
 - d) Only on an outage declared by the outage manager.
 - e) Only on an outage declared by the power station manager / project manager, where a designated manager has been Appointed.
- 23) Define supervise or supervision.
- a) The overseeing of the actions of workers to prevent accidents.
 - b) **Such a person must be trained in Risk Assessment.**
 - c) The overseeing of the actions of person(s) to such an extent as to prevent any non-dangerous act, as far as reasonably practical.
 - d) **The overseeing of actions of person(s) to such an extent as to prevent any dangerous act, as far as reasonably practical**
 - e) **Has authority to ensure precautionary measures taken are implemented**
 - f) Such a person must be trained in completion of the workers register
- 24) State the criteria required to use the Limited access register.
- a) The work must be complete within 24 hours.
 - b) **No plant isolations must be required.**
 - c) **There must be no danger to the person performing the activity, nor any risk of production loss.**
 - d) The activity must be performed by a Responsible Person.
 - e) **The activity must be completed within 12 hours.**
 - f) **The activity must be performed by a skilled person.**
- 25) State the requirements that portable electric equipment must comply with before it is used.
- a) **All Portable electrical equipment to be used during the work must be inspected by a competent person prior to commencement of work.**
 - b) **Faulty tools, cords or extensions must be withdrawn from service immediately.**
 - c) All cable joints must be done by a competent person authorised to do cable joints.
 - d) **No cords and extensions that have open wires, frayed covering on wires, cable joints, cracked plug-tops, loose pins, visible damage or faults in any way shall be used.**
 - e) Faulty tools cords or extensions must be repaired on site immediately.